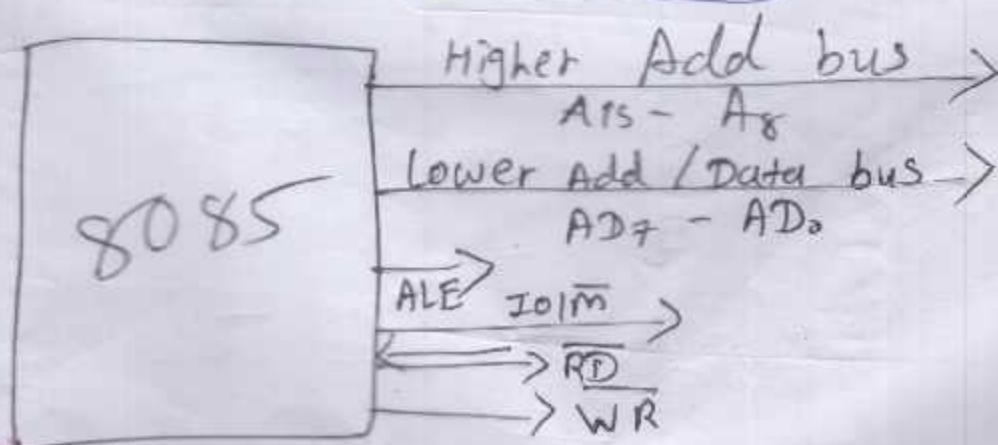
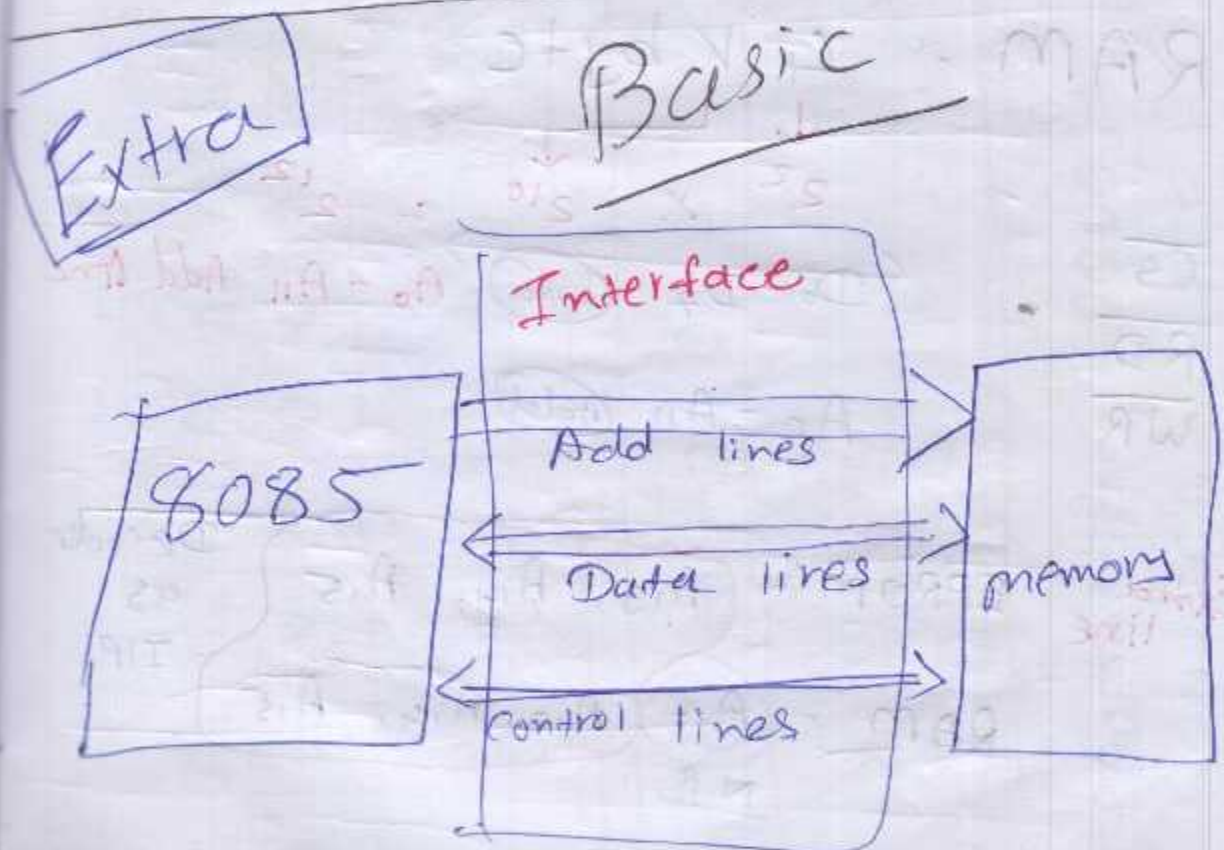


Ex 8 Kbyte EPROM & 4 Kbyte RAM
Using decoder with 8085.

- 1) 8085
- 2) EPROM - 8 Kbyte (Program memory)
- 3) RAM - 4 Kbyte (data memory)
- 4) Decoder



E.PROM - 8 k byte

$$2^3 \times 2^{10} = 2^{13}$$

$A_0 - A_{12}$ Add line

$\overline{CS}$

$D_0 - D_7$ (data)

$\overline{RD}$

$A_0 - A_{12}$ (Add)

RAM - 4 k byte

$$2^2 \times 2^{10} = 2^{12}$$

$\overline{CS}$
 $\overline{RD}$
 $\overline{WR}$

$D_0 - D_7$ (data)

$A_0 - A_{11}$ Add line

$A_0 - A_{11}$ (Add)

Extra line

E.PROM - A_{13}, A_{14}, A_{15}

Decoder as IIP

RAM - $A_{12}, A_{13}, A_{14}, A_{15}$

N.R.

memory map

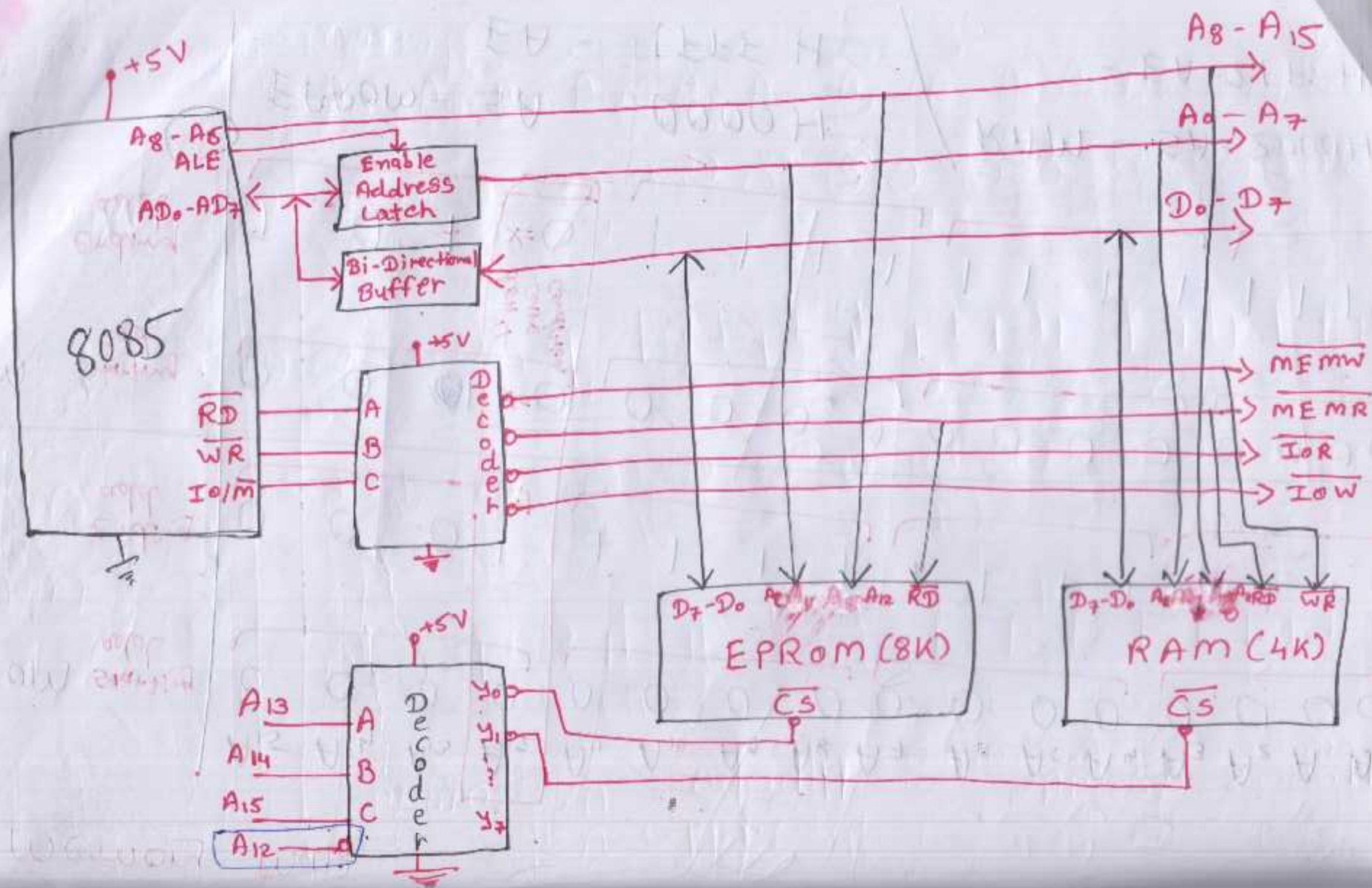
$A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$

memory map

	A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
EPROM starting add	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ending add	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1
RAM starting add	0	0	1	X=0	0	0	0	0	0	0	0	0	0	0	0	0
ending add	0	0	1	X=0	1	1	1	1	1	1	1	1	1	1	1	1

EPROM - SA - 0000 H
EA - 1FFF H

RAM - SA - 2000 H
EA - 2FFF H



Ex. Design memory system for the 8085 μP such that it should contain 8 Kbyte of EPROM & 8 Kbyte of RAM.

EPROM - 8 Kbyte

$\overline{CS}$	D ₀ -D ₇	$2^3 \times 2^{10} = 2^{13}$
$\overline{RD}$	A ₀ -A ₁₂	A ₀ - A ₁₂

RAM - 8 Kbyte

$\overline{CS}$	D ₀ -D ₇	$2^3 \times 2^{10} = 2^{13}$
$\overline{RD}$	A ₀ -A ₁₂	A ₀ - A ₁₂
$\overline{WR}$		

Extra line

A₁₃, A₁₄, A₁₅ - EPROM, RAM

Decoder I/P

Map :-

Memory

$A_{15} A_{14} A_{13} A_{12} A_{11} A_{10} A_9 A_8 A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0$

EPROM S.A

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

E.A

0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1

S.A

0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0

E.A

0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1

RAM

EPROM S.A - 0000 H

E.A - 1FFF H

RAM S.A - 2000 H

E.A - 3FFF H

Ex// Design a μP system for the 8085 μP such that it should contain 2Kbyte of EPROM & 2Kbyte of RAM with starting add $0000H$ & $6000H$ respectively.

EPROM - 2Kbyte

$\overline{CS}$ D_0-D_7
 $\overline{RD}$ A_0-A_{10}

$\downarrow$
 $2^1 \times 2^{10} = 2^{11}$
 A_0-A_{10}

RAM - 2Kbyte

$\overline{CS}$ D_0-D_7
 $\overline{RD}$ A_0-A_{10}
 $\overline{WR}$

$\downarrow$
 $2^1 \times 2^{10} = 2^{11}$
 A_0-A_{10}

Extra line

$A_{11}, A_{12}, A_{13}, A_{14}, A_{15}$

$\overline{IO/\overline{M}}$ $\overline{RD}$ $\overline{WR}$

extra for
 EPROM
 &
 RAM

because of this

3x8 decoder
 in decoder
 3 i/p required

memory map :-

From S.A

IV

RAM

$$\frac{1}{\omega_i}$$
[illegible]

EPROM - SA-0000H RAM SA-6000H
BA-07FFH BA-67FFH

* Timing Diagram

→ The graphical representation of the instruction execution in steps with respect to the time is called "Timing Diagram".

* Instruction Cycle

→ Time required to execute an instruction is called instruction cycle.

Basic UP process - Fetches
Decodes
executes } Time required for these called IC

* Machine cycle :-

→ Time required to access the memory or I/O device is called machine cycle.

→ Various operation like opcode fetch, memory read, M write, I/O Read, I/O write are performed.

* T-State :-

→ Each clock cycle is called T-states.

* 8085 Machine Cycles & their timings.

- 1) opcode fetch (4T)
- 2) memory read (3T)
- 3) memory write (3T)
- 4) I/O read (3T)
- 5) I/O write (3T)

* Required Signal

- 1) CLK
- 2) Address (Higher - $A_{15}-A_8$
Lower - A_7-A_0)
- 3) Data (D_0-D_7)
- 4) ALE
- 5) Status pins ($S_0, S_1, I/O\overline{M}$)

6) Read

7) Write

		S_0	S_1	operation
* 1 Byte 2 Byte 3 Byte } instruction		0	0	Halt
		0	1	read
		1	0	write
		1	1	opcode fetch

* Timing diagram of the

1) **MOV B, C**

→ Copy contents from source register to destination register.

→ Example:-

2000: **MOV B, C**

opcode : **MOV**

operand : **B & C**

Byte : **1 byte**

machine cycles required :- **1 (4 T-state)**

opcode fetch

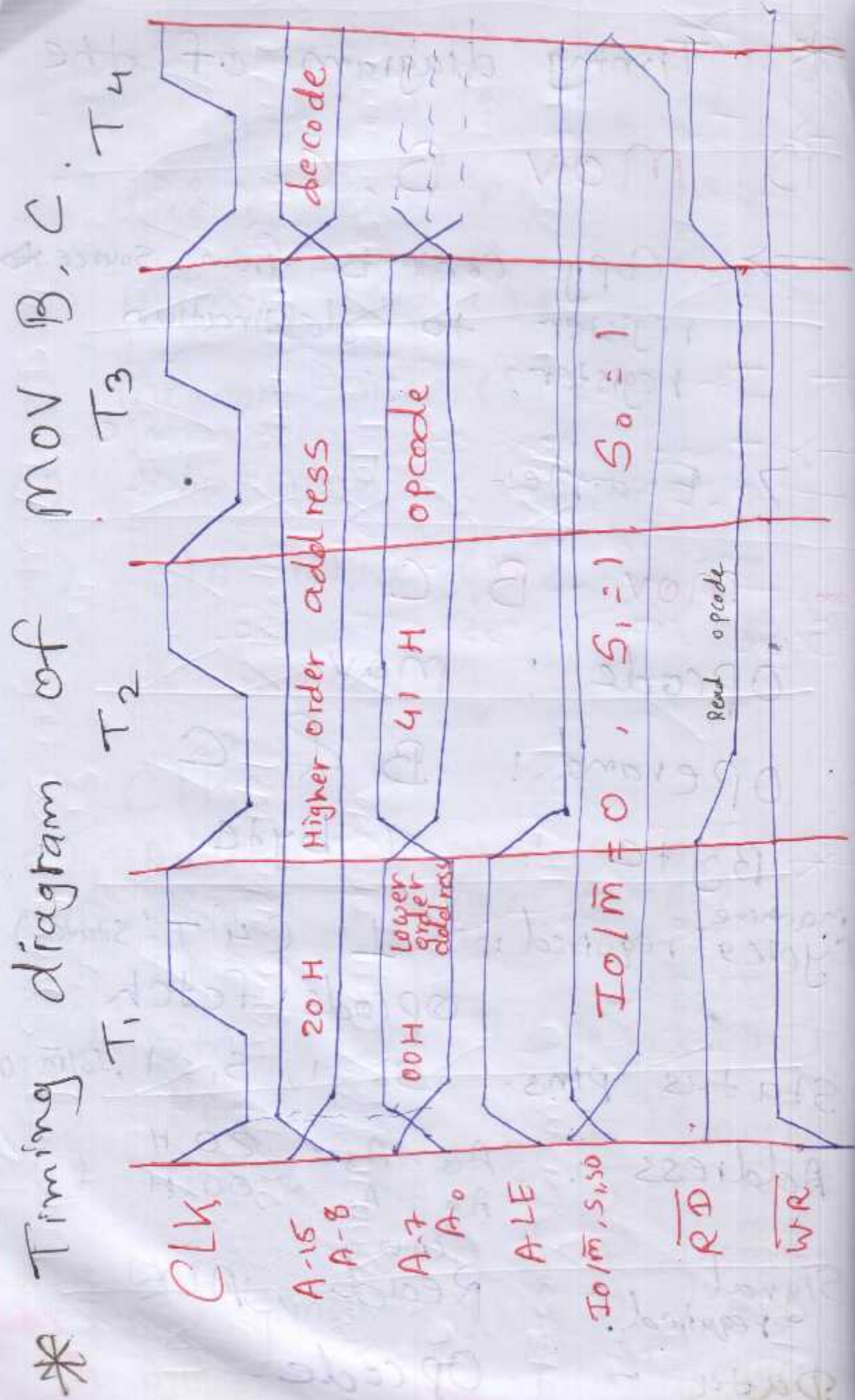
status pins - **$S_0 = 1, S_1 = 1, I_0/\bar{M} = 0$**

Address :- $A_{15} - A_8$ - **20 H**
 $A_7 - A_0$ - **00 H**

Signal required - **Read (A.L)**, **ALE**

Data - **Op code**

Fetch



* 2) $\rightarrow$ $\rightarrow$ m
opcode :
45 :
2000 :
OP
OP
By
mac
of
T-
statu
Add
signa
re

* Timing diagram of the

2) **MVI B, 45**

→ Stores the immediate
8 bit data to register or
M.L.

→ Example :-

MVI B, 45
opcode : 2000 :
45 : 2001

opcode - MVI

operand :- B

Data: 45

2000 : MVI B, 45

opcode: MVI

operand: B, 45

Byte : 2 byte

machine
cycle required: 2 1) opcode fetch (4T)
2) memory read (3T)

T-States - 7 T

Status lines - OP code - $I_o/\bar{M} = 0, S_o = 1, S_i = 1$
fetch

- memory read - $I_o/\bar{M} = 0, S_o = 0, S_i = 1$

Address - $A_{15} - A_8 = 20H$

$A_7 - A_0 = 00H$

$A_{15} - A_8 = 20H$

$A_7 - A_0 = 01H$

signal
required - Read, ALE

